Prompt 4 - neutral prompt

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two different GNN models on a compound activity prediction dataset and compare their performances. Select a compound from the test set correctly predicted as active by both models. Explain the predictions using EdgeSHAPer and determine whether the explanations are similar or different based on numerical evidence.

Assistant

I'll help you train two GNN models, compare their performances, and explain their predictions using EdgeSHAPer. Let's start by finding a suitable dataset! 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found a compound activity prediction dataset. Let me load it and set up the workflow. 📊

(Calling tool `load_dataset`)

Assistant

Perfect! I have a dataset with 8,348 compounds. Now let me split it into train/val/test partitions for GNN training. 📂

(Calling tool `split_dataset`)

Assistant

Dataset split complete! Now let me prepare the GNN dataset and train two different GNN models. I'll use **GCN** and **GAT** architectures for comparison. 🧪

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN dataset prepared. Now let me train two different GNN models: **GCN** and **GAT**. 🚀

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

Both GNN models are training in the background. Let me check their progress. ⏱️

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

Both models are still training. Let me check again after a moment. ⏳

(Calling tool `check_gnn_training`)
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Models are still training. Let me check again. ⏳

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🎉 **GCN model completed!** Training finished with impressive results:

Metric	Value
Test Accuracy	94.85%
Best Validation Accuracy	95.13%
Training Accuracy	95.34%
Matthews Correlation Coefficient	0.907

The GAT model is still training. Let me check on it. ⏳

(Calling tool `check_gnn_training`)

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GAT model is still training. Let me check again. ⏳

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(Calling tool `check_gnn_training`)

Assistant

🎉 **GAT model completed!** Both models have finished training. Here's the performance comparison:

Model	Test Accuracy	Best Val Accuracy	Training Accuracy	MCC
GCN	94.85%	95.13%	95.34%	0.907
GAT	**95.45%**	95.21%	95.38%	**0.908**

📊 **Summary**: GAT slightly outperforms GCN with a **0.6% higher test accuracy** and marginally better MCC.

Now, let me select a compound from the test set that **both models correctly predict as active**, and then explain their predictions using EdgeSHAPer. 🔬

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

Excellent! Found a compound that **both models correctly predict as active** with high confidence:

Property	Value
SMILES	`O=C(CCCc1cn(-c2ccc([N+](=O)[O-])cc2)nn1)NCCCCN1CCN(c2cccc(Cl)c2Cl)CC1`
True Label	1 (Active) ✅
GCN Prediction	1 (Active) - Confidence: 97.7%
GAT Prediction	1 (Active) - Confidence: 99.8%

Now let me load the GNN models and run **EdgeSHAPer** on this compound for both models. 🧪

(Calling tool `load_gnn_model_mcp`)
(Calling tool `load_gnn_model_mcp`)

Assistant

Both models loaded successfully! Now let me run **EdgeSHAPer** to compute edge importance for both models on this compound. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer explanations are complete for both models. Let me visualize the results and compare them. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

(Calling tool `visualize_edgeshaper_results`)

Assistant

Excellent! Now let me analyze the EdgeSHAPer results numerically and compare the explanations. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now I have all the data needed for a comprehensive comparison. Let me provide the complete analysis! 📊

📊 Complete Analysis: GCN vs GAT Performance & EdgeSHAPer Explanations

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🧪 Model Performance Comparison

Metric	GCN	GAT
Test Accuracy	94.85%	**95.45%** ✅
Best Validation Accuracy	95.13%	95.21%
Training Accuracy	95.34%	95.38%
Matthews Correlation Coefficient	0.907	**0.908** ✅
Test False Positives	102	79 ✅
Test False Negatives	147	168

🏆 **Winner: GAT** - Slightly better performance with fewer false positives!

🔬 EdgeSHAPer Explanation Analysis

Selected Compound: `O=C(CCCc1cn(-c2ccc([N+](=O)[O-])cc2)nn1)NCCCCN1CCN(c2cccc(Cl)c2Cl)CC1`

- **True Label:** 1 (Active) ✅
- **Both Models Predict:** 1 (Active)
- **GCN Confidence:** 97.7%
- **GAT Confidence:** 99.8%

📈 Numerical Comparison of Edge Importance

Top 5 Bonds by |phi_sum| (Absolute Contribution)

Rank	Bond	Structure	GCN phi_sum	GAT phi_sum	Agreement?
1	Bond 25	C-C (ring)	**0.1213**	**0.1105**	✅ Same
2	Bond 4	C-C (single)	0.1024	**0.1181**	✅ Same
3	Bond 38/36	C-N / C-C	0.1011	0.0918	⚠️ Different
4	Bond 27	N-C	0.0957	-	❌ Different
5	Bond 23/19	C-N / N-C	0.0934	0.0844	⚠️ Different

Key Numerical Evidence:

Metric	GCN	GAT
Top 2 bonds overlap	Bonds 4, 25	Bonds 4, 25
**Mean	phi	(top 10)**

Metric	GCN	GAT
Max positive phi	0.5124 (Bond 37)	0.1057 (Bond 10)
Max negative phi	-0.5241 (Bond 36)	-0.0598 (Bond 66)
Trustworthiness	0.9995	0.5675
Fidelity	0.9769	0.4983

🔍 Similarities

1. **Top 2 bonds are identical** - Both models agree the most important bonds are:
   - **Bond 4** (C-C single bond connecting the alkyl chain to the pyrazole ring)
   - **Bond 25** (C-C bond in the piperazine ring)
2. **Predominantly positive contributions** - Both models show that most top bonds contribute positively to the active prediction
3. **Similar magnitude range** - The top bonds have comparable importance (0.08-0.12 range)

⚠️ Differences

1. **Magnitude of extreme values:**
   - GCN has **much larger extreme values** (±0.52) vs GAT (±0.11)
   - This suggests GCN relies more heavily on specific bonds
2. **Different 3rd-5th ranked bonds:**
   - GCN: Bonds 38 (C-N), 27 (N-C), 23 (C-N)
   - GAT: Bonds 36 (C-C), 19 (N-C), 2 (C-C)
3. **Trustworthiness & Fidelity:**
   - GCN: **0.9995** trustworthiness (very reliable explanation)
   - GAT: **0.5675** trustworthiness (moderate reliability)

📝 Conclusion

The explanations are PARTIALLY SIMILAR but with notable differences:

Aspect	Assessment
Top 2 bonds	✅ **Highly Similar** - Both models agree
Overall pattern	✅ **Similar** - Both show positive contributions

Aspect	Assessment
Magnitude	⚠️ **Different** - GCN has more extreme values
Lower-ranked bonds	⚠️ **Different** - Different structural features
Explanation reliability	⚠️ **Different** - GCN more trustworthy

🎯 **Key Insight:** Both models agree on the **most critical structural features** (Bonds 4 and 25), suggesting these are the genuine activity-driving fragments. However, they **diverge on secondary features**, indicating different interpretations of less important structural elements. The GCN model provides more confident (higher trustworthiness) but potentially more extreme explanations, while GAT provides a more balanced but less certain explanation.



